MILITARY SPECIFICATION

STOPCOCK, SURGICAL SYRINGE, THREE WAY, AYER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a metallic three-way stopcock, of the Ayer design, suitable for use in intravenous procedures.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
PPP-B-566	Box, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-B-676	Boxes, Setup.

Military

MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWINGS

Defense Personnel Support Center (DPSC)

2026	STOPCOCK, SURGICAL SYRINGE, 3 WAY, AYER

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z70.1	American Standard Dimensions of Glass and Metal Luer Tapers for Medical Applications.

(Applications for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Material. The entire stopcock shall be fabricated from Class 8, Type 260, 270, or 360 material in accordance with GG-I-526. All surfaces shall be chromium plated over nickel plated, except the bearing surfaces of the stopcock body, rotating key and washer, and the bore through the rotating key. As an alternate, the surfaces specified to be plated shall be nickel plated only. The boil test and plating requirements specified in GG-I-526 shall apply.

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3.2 Design and construction. The item shall be a three-way stopcock, of the Ayer design, with female Luer lock connectors on two ports 90 degrees apart and with a male Luer slip connector on the third port, and shall meet all other requirements of this specification. The Luer lock connectors shall be in accordance with Figure 3 of ANSI Standard Z70.1, and the Luer slip connector shall be in accordance with Figure 2, male end, of ANSI Standard Z70.1. The stopcock shall consist of a rotating spigot that will form an easily rotatable, leak-proof system, that is integral to the stopcock body proper. The stopcock shall be in accordance with DPSC Drawing 2026.

3.3 Diameter and depth of female Luer lock cavities. The cavities of the female Luer lock connectors shall be tested for diameter and depth of Luer cavity as specified in 4.4.1. Cavities which do not fit the plug gage within the "GO" - "NO GO" limits specified in 4.4.1 shall be rejected. When the cavities are fitted onto their respective gaging members, there shall be no movement of the cavity on the gages when the connectors are pressed lightly from side to side in all directions.

3.4 Taper of male Luer slip connector. The male Luer slip connector shall be tested for proper taper as specified in 4.4.2. Male Luer slip connectors which do not fit the ring gage within the "GO - NO GO" limits specified in 4.4.2 shall be rejected. In addition, when the ring gage is fitted onto the male connector there shall be no movement of the gage on the tip when the gage is pressed lightly from side to side in all directions.

3.5 Leakage of stopcock. When tested as specified in 4.4.3, all Luer connectors and the junction of rotating spigot and stopcock body shall not show leakage exceeding the formation of one clinging drop. The rotating spigot shall not separate or loosen from the stopcock body during the leakage tests.

3.6 Clear passage of fluid path. The entire fluid path of the stopcock shall be clear and free from foreign particles to allow free passage of fluid as specified in 4.4.4.

3.7 Security of assembly. When tested as specified in 4.4.5, the rotating spigot shall not pull out of, or separate from, the stopcock body.

3.8 Identification marking. The stopcock shall be marked with the name or registered trademark of the contractor. Stopcocks of other than U.S. manufacture shall, in addition to the above, be marked with the country of origin.

3.9 Workmanship. Workmanship shall be first class throughout. Stopcocks shall be free from burrs, sharp edges, cracks, holes, pits, scale, corrosion, flaking or incomplete plating, foreign particles, and any other defects which detract from their appearance or impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the copper alloys may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number (NSN), the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in Table I. The unit of product for examination purposes shall be one stopcock.

Table I. Sampling for examination

Type defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table II. The unit of product for test purposes shall be one stopcock. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

Table II. Sampling for tests

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Stopcock	Boil test	3.1	GG-I-526	S-2	1.5
	Plating test	3.1	GG-I-526	S-2	1.5
	Dimensions	3.2	-	S-2	2.5
	Diameter and depth of female Luer cavities	3.3	4.4.1	S-4	1.0
	Taper of male Luer connector	3.4	4.4.2	S-4	1.0
	Leakage test	3.5	4.4.3	S-4	1.0
	Clear passage of fluid path	3.6	4.4.4	S-2	1.5
	Security of assembly	3.7	4.4.5	S-2	1.5

4.2.3 Raw materials. A sufficient quantity of each lot of each metal used in the construction of the stopcock shall be selected for chemical analysis.

4.3 Examination.

4.3.1 Examination of stopcock. The stopcock shall be examined to determine compliance with requirements contained in this specification. Examination shall be conducted as specified in Tables I and III. Examination, however, shall not be restricted to defects listed in Table III. Any stopcocks in the sample containing one or more defects shall be rejected and if the number of defective stopcocks in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

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Table III. Classification of defects

Categories	Defects
Critical	None defined.
Major	
101	Stopcock not of design specified.
102	Rotating spigot (or key) does not exhibit smooth operation without sticking or binding in any position.
103	Stopcock not plated as specified.
104	Plating not uniform.
105	Plating not free from flaking.
106	Fluid paths not free from foreign matter or obstructions.
107	Stopcock not free from burrs, sharp edges, cracks, holes, pits, scale or corrosion.
108	Rotating spigot not secured to stopcock body (i.e. separates from body when rotated).
Minor	
201	Stopcock not identified as specified in 3.8.

4.3.2 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The Inspection Level shall be S-2 and the Acceptable Quality Level shall be 4.0 defects per hundred units.

Table IV. Examination of preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

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4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Diameter and depth of female Luer lock cavities. The Luer cavities of the female connectors shall be tested with the male Luer taper plug gage in accordance with Figure 5 of ANSI Standard Z70.1 (this gage shall be as shown in the ANSI Standard, except that it shall be constructed with a suitable fluid path to permit performance of the leakage test specified in 4.4.3). The female connectors shall be placed onto this gage with a straight motion, without turning or twisting the plug gage into the cavity. Once the female Luer cavity is placed onto the gage, no additional force shall be exerted to force the plug gage into the Luer cavity. The fit of the Luer cavities onto this gage shall be tested by means of the feeler gage in accordance with Figures 5 and 6 of ANSI Standard Z70.1. The cavity shall be pressed lightly from side to side in all directions on the gage. The connector shall meet the requirements of 3.3.

4.4.2 Taper of male Luer slip connector. The male connector shall be tested with the female Luer ring gage in accordance with Figure 7 of ANSI Standard Z70.1. The male connector shall be placed into this gage with a straight motion, without turning or twisting the male connector into the gage. Once the male connector is placed into the gage, no additional force shall be exerted to force the male connector into the gage. The gage shall be pressed lightly from side to side in all directions on the connector. The connector shall meet the requirements of 3.4.

4.4.3 Leakage test. The stopcock shall be subjected to an internal water pressure of 45 pounds per square inch gage (p.s.i.g.). The plug gage specified in 4.4.1 shall be inserted into the female Luer cavity (the female cavity in line with the male connector) with a straight motion, without turning or twisting the plug gage into the cavity with a force not to exceed 5 pounds. The feeler gage shall be placed onto this plug gage to assure that the "GO" position is not violated. Another plug gage (not modified with a fluid path) in accordance with Figure 5 of ANSI Standard Z70.1 shall be inserted into the other female Luer cavity with a straight motion, without turning or twisting the plug gage into the cavity with a force not to exceed 5 pounds. The feeler gage shall be placed onto this plug gage to again assure that the "GO" position is not violated. A hypodermic needle, with the lumen sealed, shall be used to seal the male connector in the following manner: The hub of the hypodermic needle shall be tested to comply with suitable cavity as specified in 4.4.1. A needle possessing a hub cavity that passes the diameter and depth of Luer cavity shall then be used to seal the male connector of the stopcock. Water shall then be internally admitted to the testing arrangement in a leakproof manner until each fluid path is filled (air bubbles eliminated) and tested, in turn, with internal pressure of 45 p.s.i.g. applied for 30 seconds. The entire assembly shall be noted for leakage and spigot performance as specified in 3.5.

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4.4.4 Clear passage of fluid path. The stopcock shall be tested with an appropriate testing fluid applied internally to the stopcock with the stopcock placed in a horizontal position. Each path shall be tested, in turn, using testing assembly fixtures as specified in 4.4.3, except that one port shall be opened to atmosphere for fluid exit during the test for that fluid path. The stopcock shall allow free passage of the testing fluid within 5 seconds after an internal pressure of 3 ± 1/2 p.s.i.g. has been applied to the stopcock. One liter of testing fluid shall be passed through the stopcock and then through filter paper. The filter paper shall be examined for presence of foreign matter.

4.4.5 Security of assembly. The stopcock shall be tested for security of assembly by applying a straight force or pull of 10 pounds to the rotating spigot member for a period of at least 1 minute with the stopcock body appropriately secured.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. Each stopcock shall be packaged in a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676. Closure shall be adequate to prevent accidental opening.

5.1.1.2 Intermediate package. Twelve (12) units of issue shall be packaged in a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676. Closure shall be adequate to prevent accidental opening.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Level A. Two hundred and eighty eight (288) units of issue shall be packed in a box designed for a Type 2 load and constructed in accordance with PPP-B-585, Class 3, Style 3; PPP-B-601, overseas type; PPP-B-621, Class 2 or PPP-B-636, Type CF, Class weather-resistant. Closure and strapping shall be in accordance with the appendix of the applicable box specification.

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5.2.1.1 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard boxes shall be waterproofed in accordance with PPP-B-636.

5.2.2 Level B. Two hundred and eighty eight (288) units of issue shall be packed in a box designed for a Type 2 load, and constructed in accordance with PPP-B-585, Class 2, Style 3; PPP-B-601, domestic type; PPP-B-621, Class 1; or PPP-B-636, Type CF, Class domestic. Closure shall be as specified in the appendix of the applicable box specification. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The stopcock covered by this specification is intended to be used in intravenous procedures.

6.2 Ordering data. Procurement documents should specify:

(a) Title, number and date of this specification.
(b) National Stock Number (NSN)
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.3 This specification is intended to cover the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6515-00-374-4800	STOPCOCK, SURGICAL SYRINGE, THREE WAY, AYER

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

Project No. 6515-1900
cs

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-S-37122 STOPCOCK, SURGICAL SYRINGE, THREE WAY, AYER

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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